Tree Canopy Health and Urban Vulnerability in NYC

Columbia GSAPP Intro to Urban Informatics Fall 2025

Talya Soytas UNI: ts3651

New York City is working on a comprehensive Urban Forest Plan which will aim to increase the canopy cover from 23% to 30% by 2030, which is a lofty goal, but one that requires studying the issues with the current canopy to make informed decisions as well (New York City Urban Forest Plan, n.d.). Streets in particular are essential spaces to consider for canopy cover increase, as most of the city is quite densely developed. But before expansions, it is important to look at how the existing canopy is faring. While the upcoming plan wants to prioritize underserved neighborhoods, typically home to black and brown and low income communities, there is also tension with impervious surface cover. Impervious surfaces were found as a negative predictor of poorer tree health (Just et al., 2018), thus may drive decisions against tree plantings in these areas. This study examines the NYC specific context for these social and built environment vulnerabilities in conflict for tree planting and maintenance to inform programming decisions for the upcoming Urban Forest Plan.

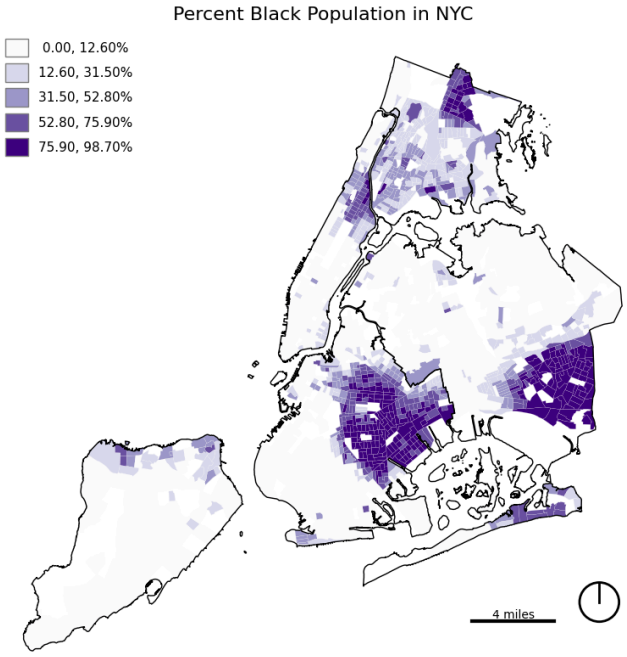
Data and Scope

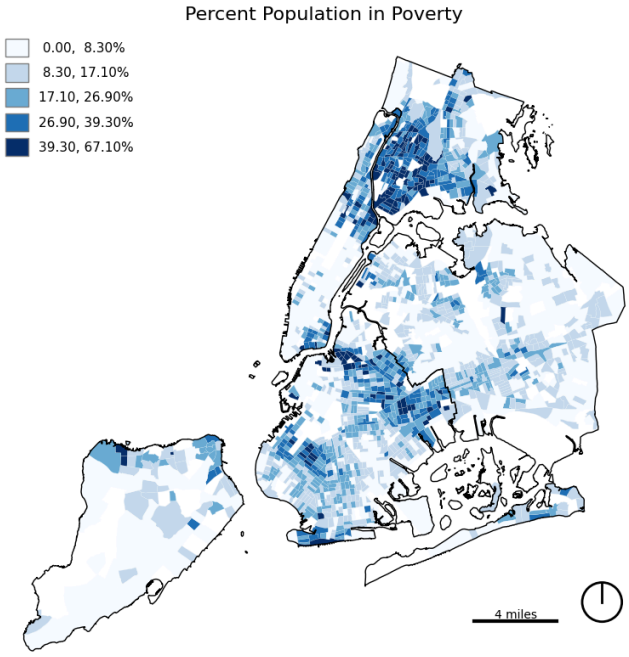
The analysis combines NYC's 2015 Street Tree Census, tract-level demographic data from the American Community Survey (ACS 2015), and the National Land Cover Database (NLCD 2015) impervious surface raster. Tree health was categorized into "Good," "Fair," "Poor," or "Dead." Demographic variables included poverty, race/ethnicity, and limited English proficiency, aggregated to the census tract level. Impervious surface fractions were computed per tract from a raster to capture the intensity of urban infrastructure.

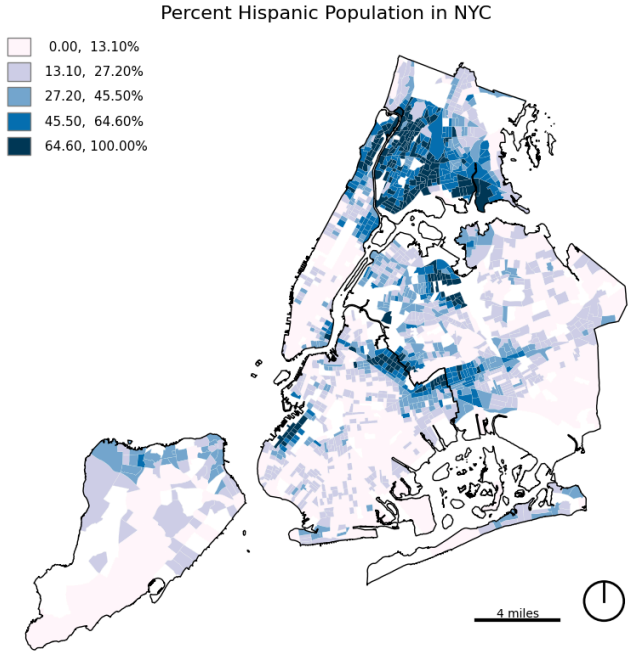
Questions

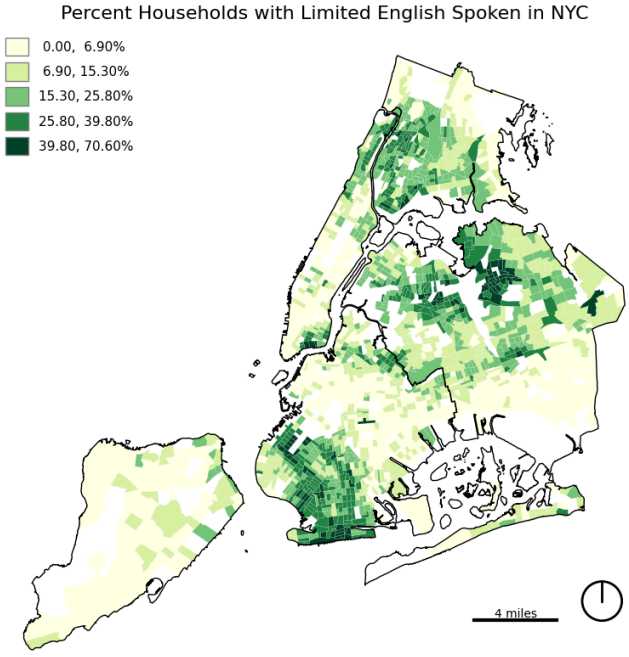
- According to the last tree census (2015), what areas of NYC had street trees in poor condition?
    - How do these areas compare to high social vulnerability neighborhoods in NYC?
    - How do these areas compare to neighborhoods with a high proportion of impervious surfaces?

Methods and Key Findings

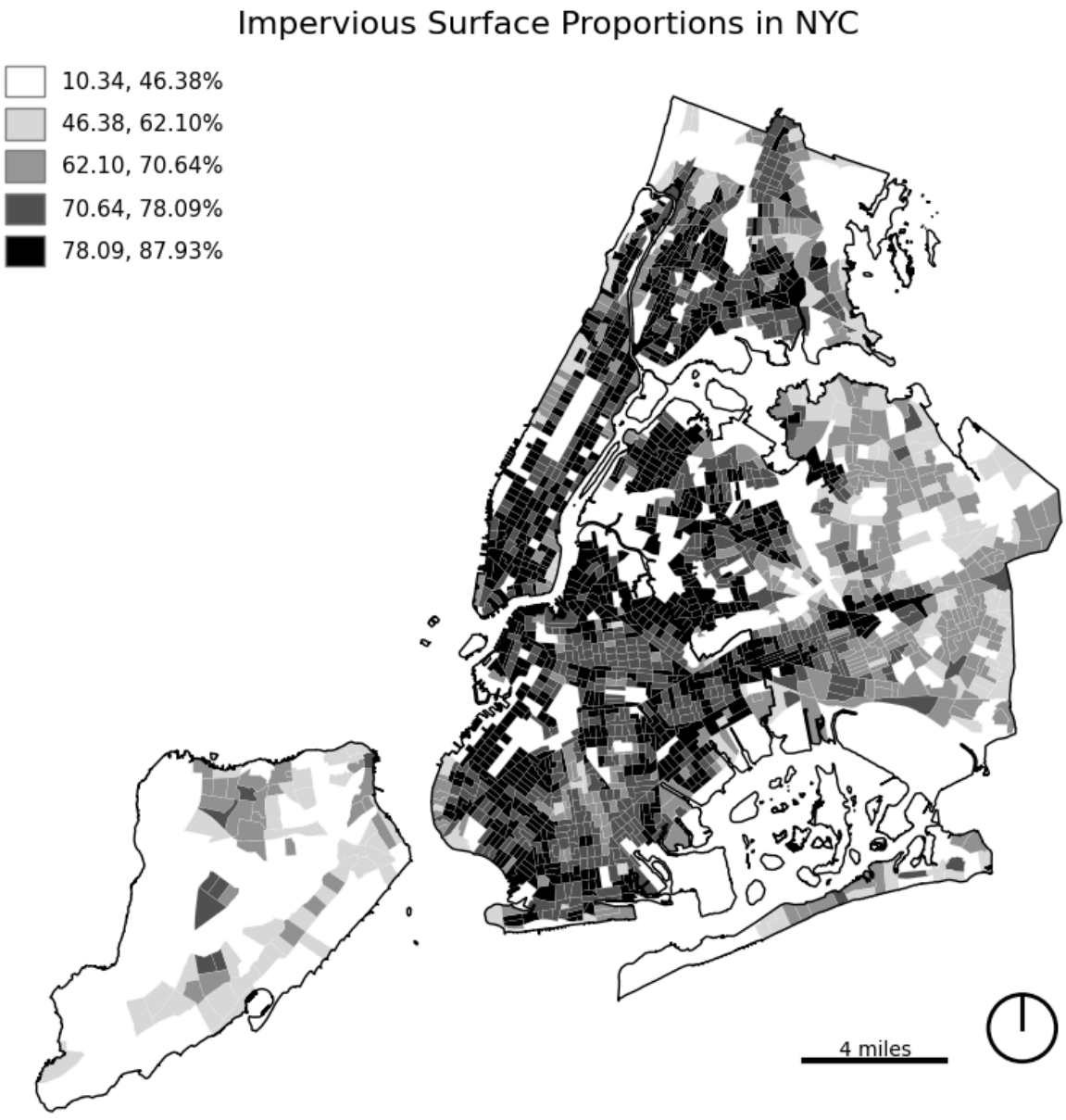




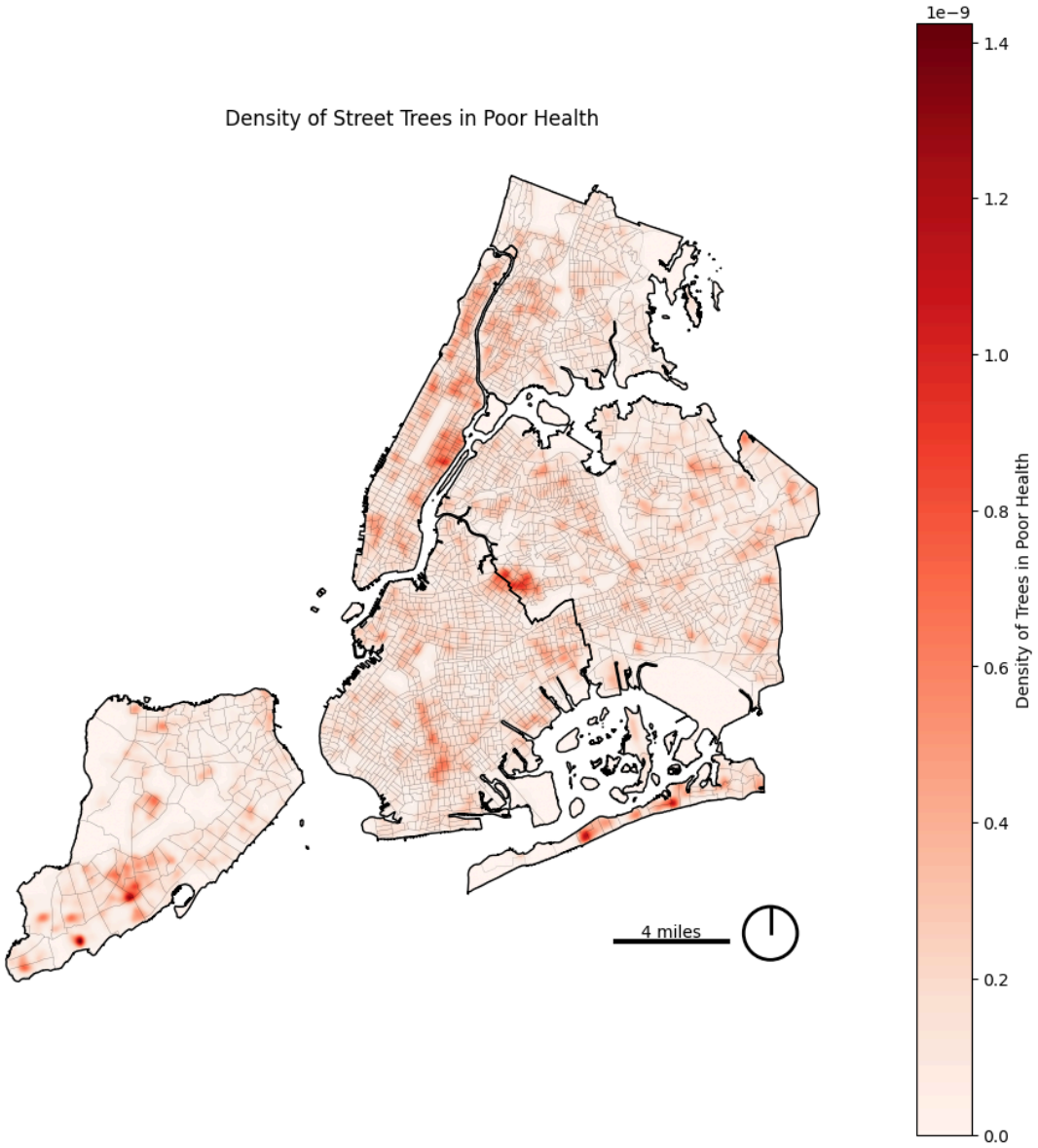




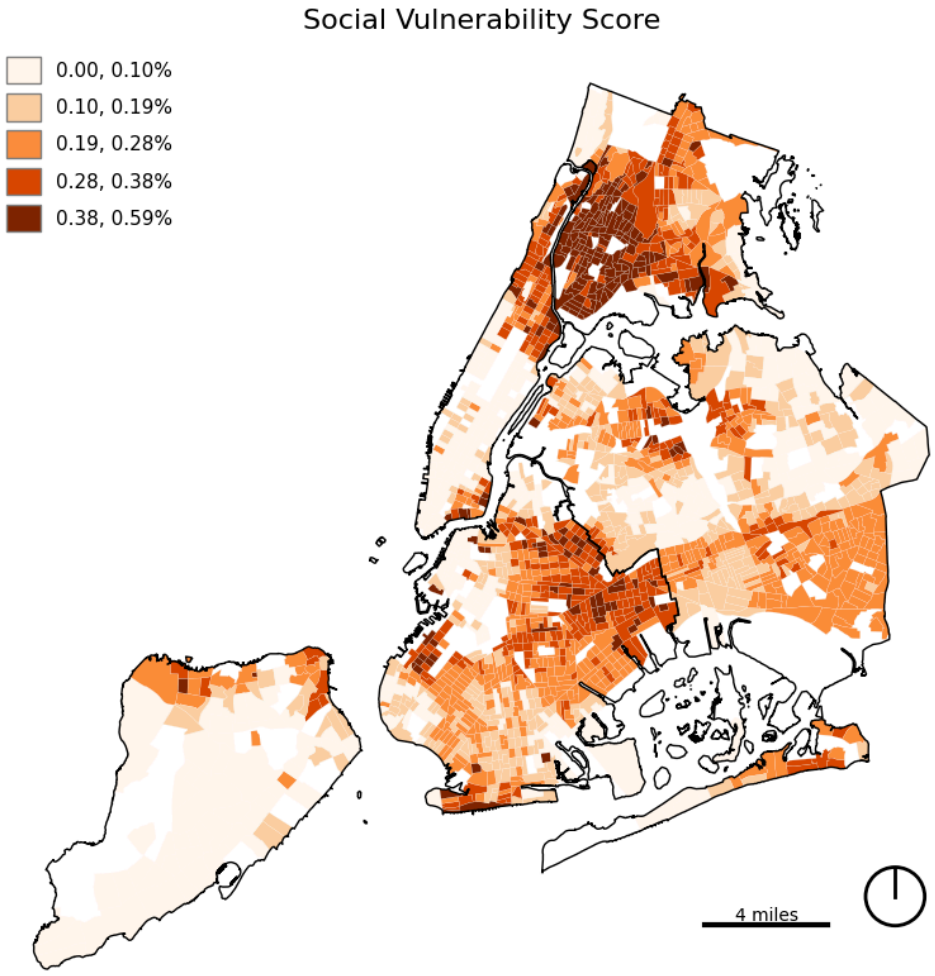
First, ACS social vulnerability parameters were spatialized for NYC from the 2015 survey.



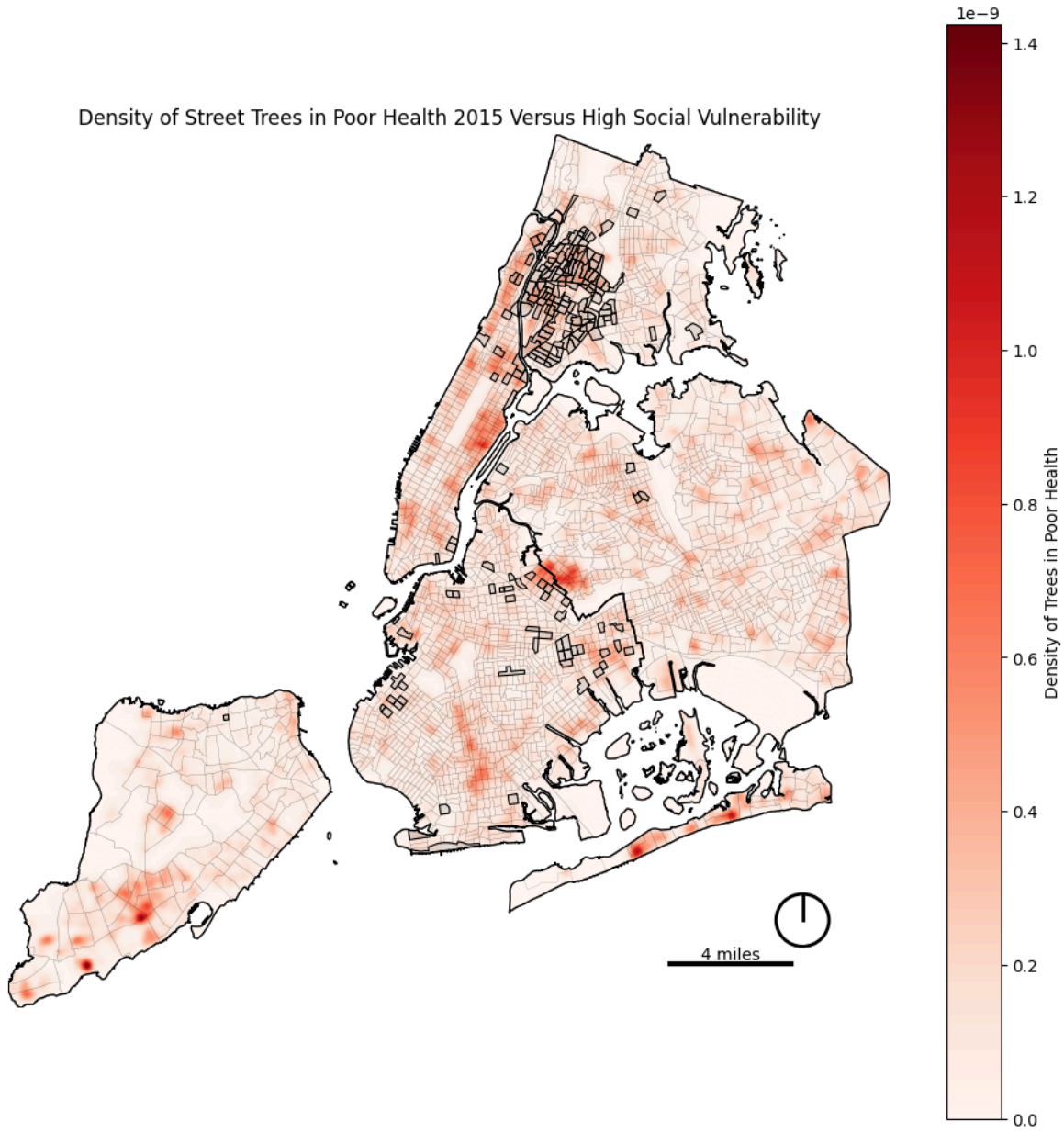
A look at the 2015 impervious surface percentages across the city revealed areas of dense urbanisation, lack of permeable, green spaces, and also coincided with some industrial zones like Bushwick and Ridgewood areas.



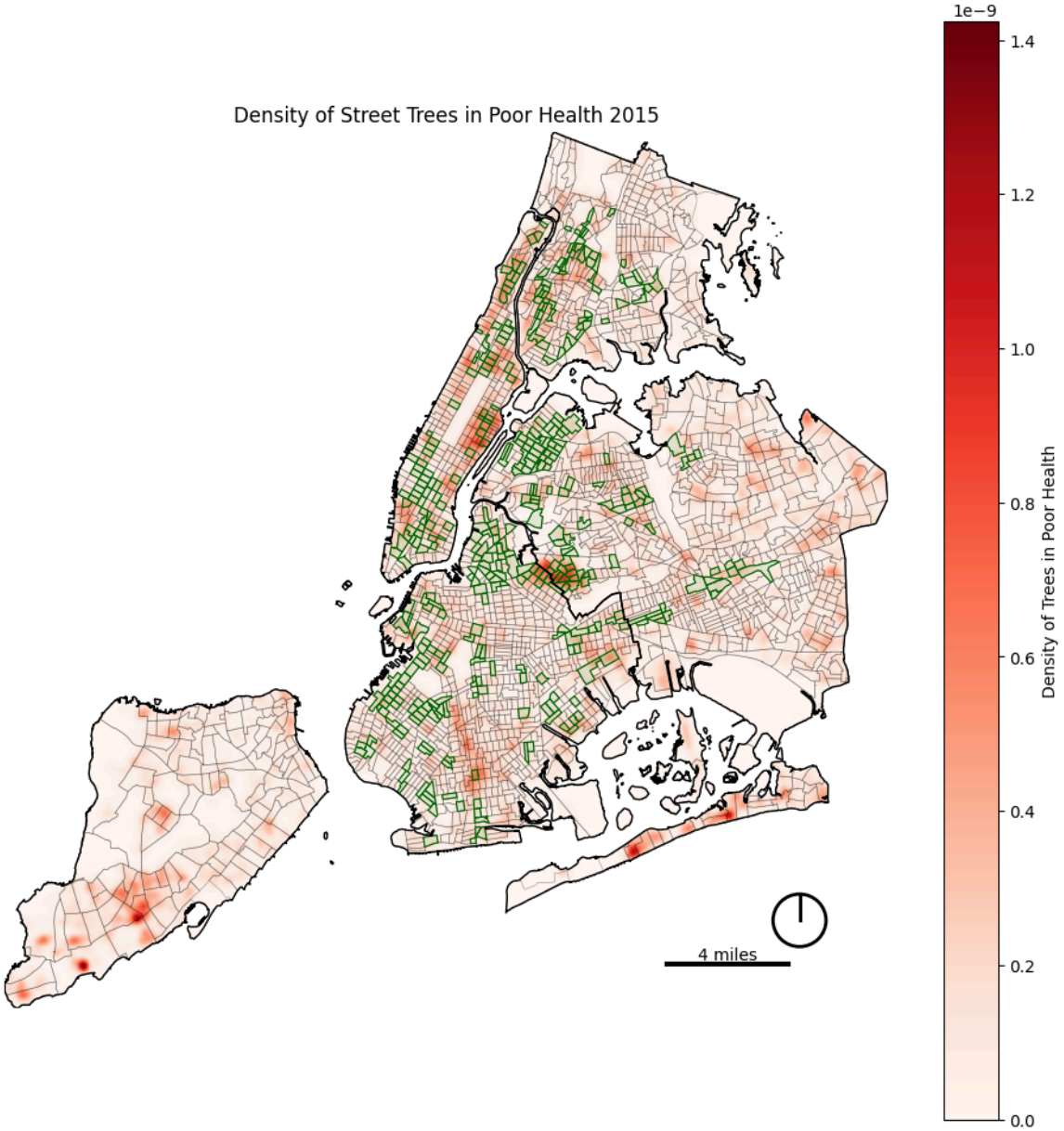
Then the concentration of street trees in poor health was mapped out using Kernel Density Estimation on the 2015 Street Tree Census Data. This map reveals some hotspots in Bushwick, Ridgewood, Upper East Side, and the Rockaways, while North Manhattan, Central Brooklyn and the Bronx showed moderate counts of street trees in poor health.



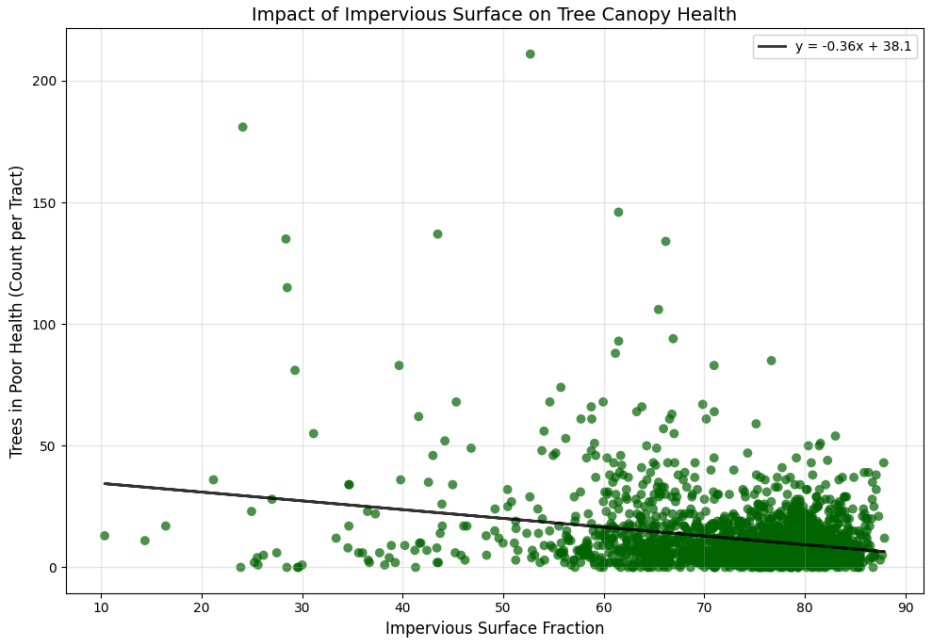
To compare this density to demographics, the vulnerability variables were turned into an index through the MCDA process, which resulted in a mapping of historical social inequities and vulnerabilities in terms of access to greenspace and canopy care services. Parts of Brooklyn (including Bushwick), Upper Manhattan, and most notably the Bronx stood out as the vulnerable areas. These are coincidently areas that are flagged as highly impervious as well.



When the top 10% highest vulnerability tracts were overlaid with the poor tree health however, there wasn't an immediately apparent overlap of poor canopy hotspots and vulnerability criteria.



Then, the same comparison was made with impervious areas proportion in NYC, which presented some overlap (like the literature suggested), the top 20% of the impervious areas. Specifically in the Upper East Side and Bushwick clusters of poor canopy health, this may be a strong predictor of the poor canopy health.



A scatterplot and regression were used to further evaluate the relationship between impervious fraction and counts of trees in poor health, which doesn't show an overall positive correlation of variables. Which means the two overlap areas may be one-off, and that local context is highly relevant to street tree health.

Limitations

- Impervious surface fraction serves as a proxy for stress, but other factors (soil quality, microclimate, species selection etc.) also influence canopy health.
- The MCDA approach reflects relative vulnerability and should be interpreted as indicative rather than definitive.
- Due to time limitations, a more comprehensive look at canopy health through time was not conducted, but analyzing the 1995, 2005, and 2015 tree census may shine a light to neighborhoods that had consistently poor tree health over the years for more concrete takeaways.
-

Conclusion

The upcoming Urban Forest Plan will attempt to create a justice framework for tree planting and maintenance, allocating more resources to disadvantaged communities. However, said communities are also in areas of high imperviousness in the city, which has been tied to poor canopy health. Thus, the plan will need to consider planting and maintenance strategies that address the challenges of building an urban forest in a mostly paved landscape, whether that be through different technologies, species selection, or maintenance guidelines taking this tough terrain into consideration.

Resources

New York City Urban Forest Plan. (n.d.). *Why trees?* Retrieved October 19, 2025, from https://www.urbanforestplan.nyc/#why-trees

Just, M. G., Dale, A. G., Frank, S. D., & Youngsteadt, E. (2018). *Impervious surface thresholds for urban tree site selection. Urban Forestry & Urban Greening, 32,* 191–198.

Data Sources

New York City Department of City Planning. (2024). *Borough Boundaries* [Data set]. NYC Open Data. https://data.cityofnewyork.us/d/gthc-hcne

New York City Department of Parks & Recreation. (2016). *2015 Street Tree Census – Tree Data* [Data set]. NYC Open Data. https://data.cityofnewyork.us/Environment/2015-Street-Tree-Census-Tree-Data/uvpi-gqnh

New York City Department of Transportation. (n.d.). *Automated Traffic Volume Counts (2015 subset)* [Data set]. NYC Open Data. https://data.cityofnewyork.us/d/7ym2-wayt

U.S. Census Bureau. (2015). *ACS demographic and housing estimates (Table DP05), American Community Survey 1-year estimates — New York City counties subset* [Data set]. https://data.census.gov/table/ACSDP1Y2015.DP05

U.S. Census Bureau. (2015). *Poverty Status of Families in the Last 12 Months (Table S1702), American Community Survey 1-year estimates — New York City counties subset* [Data set]. https://data.census.gov/table?q=S1702&y=2015

U.S. Census Bureau. (2015). *Limited English Speaking Households (Table S1602), American Community Survey 1-year estimates — New York City counties subset* [Data set]. https://data.census.gov/table?q=S1602&y=2015

U.S. Census Bureau. (2015). *Educational Attainment (Table S1501), American Community Survey 1-year estimates — New York City counties subset* [Data set]. https://data.census.gov/table?q=S1501&y=2015

U.S. Geological Survey. (2024). *Annual NLCD Collection 1: Fractional Impervious Surface 2015* [Data set]. MRLC. https://www.mrlc.gov/downloads/sciweb1/shared/mrlc/metadata/Annual_NLCD_FctImp_2015_CU_C1V1.xml